Global Olympic Sports Performance Analysis: Project Documentation

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1. Project Overview

This project analyses historical Olympic Games data in Tableau to show who participates, who wins medals, and which sports and countries perform best.

Objective: build an interactive dashboard and three data stories, then publish them on a website hosted with GitHub Pages.

Questions answered:

- What age, height and weight do medal winners have?
- Which sports and countries win the most medals?
- How has male and female participation changed over the years?
- How do Summer and Winter Games compare in size?

2. Tools and Technologies

Tool	Purpose
Tableau Desktop	Data connection, preparation, charts, dashboard and stories
Tableau Public	Publishing the dashboard and stories online
HTML and CSS	Building the project website
GitHub	Storing project files and version history
GitHub Pages	Hosting the website for free
Skill Wallet	Task tracking (Kanban) and project submission

3. Dataset and Data Collection

The dataset records Olympic athletes and their results. The dashboard uses these fields:

- **Athlete details:** age, sex, height, weight

- **Event details:** sport, year, season (Summer or Winter)
- **Result details:** country and medal (Gold, Silver, Bronze)

Source: Olympic athlete event results dataset (athletes, sports, countries and medals), included in the Tableau workbook in this repository.

Size: The full dataset is loaded in Tableau, and all charts are built from it.

4. Step-by-Step Process

1. **Collect the dataset.** Download the Olympic athlete dataset and check its columns.
2. **Connect to Tableau.** Open Tableau Desktop and connect to the data file. Check that each field has the right data type.
3. **Prepare the data.** Remove or fix missing and duplicate values, and set fields as dimensions or measures.
4. **Create the visualizations.** Build one sheet for each chart:
   - Age, weight and height distribution of winners (histograms coloured by medal)
   - Top 10 sports by total medals
   - Top 10 countries by medals
   - Male vs female participation over the years
   - Weight and height correlation (scatter plot)
   - Total athletes by year and season
5. **Build the dashboard.** Arrange the eight charts in a 3 by 3 layout with the Olympic rings logo in the centre.
6. **Create the stories.** Build three Tableau Stories, each with scenes and captions that explain the findings.
7. **Test performance and layout.** Check the filters, the data load time and the dashboard size (see section 5).
8. **Publish.** Publish the workbook to Tableau Public.
9. **Embed in a website.** Add the dashboard and stories to an HTML page and host it on GitHub Pages (see section 6).

5. Performance Testing and Responsive Design

The dashboard has 8 charts built from 4 chart types: histograms, horizontal bars, lines and a scatter plot. The table lists the checks recorded for this project.

Check	Result
Number of visualizations on the dashboard	8 charts plus the logo
Unique chart types	4 (histogram, bar, line, scatter)
Stories created	3
Publishing	Tableau Public, embedded in a GitHub Pages website

Responsive design: the website uses a fluid layout that adapts to laptop and phone screens, and the Tableau views resize inside their container.

6. Website Integration and Publishing

The website is a single `index.html` page that embeds the published Tableau dashboard and the three stories using the Tableau Embedding API. It has an overview, the live dashboard, the stories, key insights and the tools used.

1. Create a public GitHub repository for the project.
2. Upload `index.html`, the dashboard PDF and the Tableau workbook (.twbx).
3. Open Settings, then Pages, and set the source to the `main` branch and the root folder.
4. Wait one to two minutes for GitHub Pages to deploy.
5. Open the live site and check that the dashboard and all three stories load.
6. Add all links to the README file.

7. Key Insights

- **Winner profile:** most medal winners are in their mid-20s to early 30s, with weight mostly between 60 and 85 kg and height mostly between 170 and 185 cm.
- **Top sports:** Athletics leads with about 3,969 medals. Swimming and Rowing (about 2,945) follow, then Gymnastics, Fencing and Cycling.
- **Top countries:** the USA leads with about 5,637 medals, ahead of the former USSR, Germany and Great Britain.
- **Participation:** total athletes grew strongly over time, and female participation rose sharply after about 1960.
- **Seasons:** Summer Games have far more athletes than Winter Games.

8. Project Links and Conclusion

Item	Link
GitHub repository	Global-Olympic-Sports-Performance-Analysis
Live website	GitHub Pages site
Tableau dashboard	Dashboard 1
Story 1	Story 1
Story 2	Story 2
Story 3	Story 3
Demo video	Demo video (Google Drive folder)

Conclusion: the analysis shows that Olympic participation has grown steadily, that a small group of sports and countries wins most medals, and that female participation has risen sharply. Tableau made these patterns easy to explore, and the website lets anyone view them online.